# Teacher answer key: Primary 5, Semester 2 (DRAFT)

**DRAFT: this answer key is for owner review. Do not give it to pupils. Contact: pypteltd@gmail.com (business email for now, to be changed later)**

This key lists the answers and the result of the check on every page. You do not need it for routine marking: a pupil who has a working check has the right answers. To spot-check, compare the check strip with the line "Check result" below.

### How the topic tags work

Each entry names the syllabus outcomes the puzzle practises, using the outcome numbers of the Primary 5 (Standard) content in the Mathematics Syllabus, Primary (2021 edition, pages 41 and 42) where our summary prints them. The wording of each outcome line is our own summary, not the syllabus text. The references are for your planning only and are not printed on the pupil pages. "Problem solving" marks a page whose questions come from the kinds found in school tests rather than from one numbered outcome.

### Pages in this booklet (Terms 3 and 4)

| Strand | Pages |
|---|---|
| Number and Algebra - Percentage | S2-01, S2-02, S2-03, S2-04, S2-05 |
| Number and Algebra - Rate | S2-06, S2-07 |
| Measurement and Geometry - Area and Volume | S2-08, S2-09, S2-10, S2-11, S2-12, S2-13, S2-14, S2-15 |
| Measurement and Geometry - Geometry | S2-16, S2-17, S2-18, S2-19, S2-20 |

The page code in the footer, for example S2-07 (P5-27), gives the page in this booklet and, in brackets, its number in the full set of 40.

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## S2-01 (P5-21) How Many Percent? 1 star

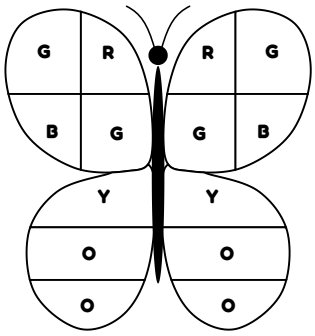
**Topic: Percentage: a part of a whole as a percentage**

Syllabus outcomes: Percentage: a part of a whole as a percentage. Notes on the tag: expressing a part of a whole as a percentage; the sign %
Puzzle type: Code-breaker (percentages). How the pupil checks: Code word (6 letters).
**Answers:** 1) 10%; 2) 40%; 3) 5%; 4) 30%; 5) 75%; 6) 98%
**Check result:** Code word: FRIDGE
**Teacher note:** Six questions: four bars divided into equal parts with some parts shaded, and two stories with a part and a whole. The pupil writes the part of the whole as a percentage. The decoy letters stand for the usual slips: giving the number of shaded parts, giving the percentage not shaded, or giving the number of parts left over. A box left blank gives no letter and the check cannot see it.

## S2-02 (P5-22) Colour the Butterfly 1 star

**Topic: Percentage: a part of a whole, a percentage of a quantity, discount and GST**

Syllabus outcomes: Percentage: a part of a whole as a percentage, Percentage: a percentage part of a whole, Percentage 1.4. Notes on the tag: a percentage part of a whole, a part of a whole as a percentage, discount and GST (percentage)
Puzzle type: Colour by code (pupils colour with their own crayons). How the pupil checks: Colour key and tally: every answer must be on the key, the number of parts of each colour must match the box, and the parts in the same place on the left and on the right match.
**Answers:** Colour key: 12 = yellow, 20 = red, 25 = blue, 40 = green, 60 = orange
**Check result:** Tally: yellow 2, red 2, blue 2, green 4, orange 4; the parts in the same place on the left and on the right match.




**Teacher note:** Every part asks for a percentage of a quantity, a part of a whole written as a percentage, a price after a discount, or a price after adding a percentage such as GST. Pupils colour the black-and-white outline with their own crayons (or, for a pencil-only copy, write Y, R, B, G or O in each part: the tally and the matching parts can be checked the same way). Check strip: count the parts of each colour (the tally is printed); the parts in the same place on the left and on the right match.

## S2-03 (P5-23) Sale Prices 2 stars

**Topic: Percentage: finding the discount and the sale price**

Syllabus outcomes: Percentage 1.4, Percentage: a percentage part of a whole. Notes on the tag: finding discount and the price after discount
Puzzle type: Ordering (lettered cards). How the pupil checks: Letters of the cards in the order asked spell a word (6 letters).
**Answers:** Order (smallest sale price first): Price: $30 Discount: 50% -> $15, Price: $30 Discount: 25% -> $22.50, Price: $30 Discount: 15% -> $25.50, Price: $100 Discount: 40% -> $60, Price: $250 Discount: 20% -> $200, Price: $250 Discount: 10% -> $225
**Check result:** Word: THIRDS
**Teacher note:** Six price tags, each with an original price and a percentage discount. The pupil finds the discount, takes it off the price and puts the six sale prices in order from smallest to greatest. The cards are not in order of sale price. The letters, read in that order, spell a word with exactly one arrangement that is a word, so no wrong order spells a word. A box left blank gives no letter and the check cannot see it.

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## S2-04 (P5-24) GST Match  2 stars

**Topic: Percentage: GST**

Syllabus outcomes: Percentage 1.4, Percentage: a percentage part of a whole. Notes on the tag: finding GST and the price with GST
Puzzle type: Matching (lettered cards). How the pupil checks: Letters of the matching cards, tile 1 to 6, spell a word.
**Answers:** W: Before GST: $45 -> With GST: $49.05; A: Before GST: $55 -> With GST: $59.95; L: Before GST: $60 -> With GST: $65.40; R: Before GST: $155 -> With GST: $168.95; U: Before GST: $160 -> With GST: $174.40; S: Before GST: $165 -> With GST: $179.85
**Check result:** Word: WALRUS
**Teacher note:** Six cards with a price before GST (multiples of $5) and six tiles with the price with 9% GST, in order of size. The pupil finds 9% of the price, adds it, and matches the card with the tile. The letters, read from tile 1 to tile 6, spell a word with exactly one arrangement that is a word, so no wrong matching spells a word. GST is 9%; the page states the rate. A box left blank gives no letter and the check cannot see it.

## S2-05 (P5-25) Annual Interest  2 stars

**Topic: Percentage: annual interest**

Syllabus outcomes: Percentage 1.4, Percentage: a percentage part of a whole (problem solving). Notes on the tag: finding annual interest
Puzzle type: Code-breaker (interest in money stories). How the pupil checks: Code word (6 letters).
**Answers:** 1) $150; 2) $54; 3) $1260; 4) $8; 5) $50; 6) $530
**Check result:** Code word: STUDIO
**Teacher note:** Six savings stories with a percentage interest paid at the end of each year, in whole dollars. The interest is a percentage of the first amount saved and is the same every year (the page says so); the pupil finds the interest for one year or several, or the money in the bank. The decoy letters stand for the usual slips: giving the interest when the total is asked, the total when the interest is asked, or the interest for one year too many or too few. Interest on interest (compound interest) is not used. A box left blank gives no letter and the check cannot see it.

## S2-06 (P5-26) Rate Cards  1 star

**Topic: Rate: finding the rate, the total or the number of units**

Syllabus outcomes: Rate (problem solving). Notes on the tag: rate as the amount of one quantity per unit of another quantity; finding the rate, the total or the number of units
Puzzle type: Lettered choices (rate). How the pupil checks: Word from the letters of the right choices (6 letters).
**Answers:** 1) 5 ℓ; 2) $9.60; 3) 6 hours; 4) 12 boxes; 5) 48 ℓ; 6) 3 kg
**Check result:** Word: STRONG
**Teacher note:** Six one-step rate questions: two ask for the rate (an amount for one unit), two for the total at a given rate and two for the number of units. The wrong choices are the usual slips (dividing the wrong way round, multiplying instead of dividing, adding or subtracting the two numbers). Each choice has a letter and the right choices spell the word. Every other way of choosing was tested against a very large word list. A box left blank gives no letter and the check cannot see it.

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## S2-07 (P5-27) Rate Stories  2 stars

**Topic: Rate: two-step problems**

Syllabus outcomes: Rate (problem solving). Notes on the tag: rate: finding the rate, then the total or the number of units
Puzzle type: Code-breaker (rate stories). How the pupil checks: Code word (6 letters).
**Answers:** 1) 30ℓ; 2) 45min; 3) 54 dollars; 4) 72bolts; 5) 24min; 6) 70 dollars
**Check result:** Code word: WISDOM
**Teacher note:** Six two-step rate stories: the pupil finds the rate for one unit (litres in one minute, price of one kilogram, pages in one minute) and then the total for a stated number of units, or the number of units for a stated total. All answers are whole numbers. The decoy letters stand for the usual slips: the rate turned upside down, or only the rate given. A box left blank gives no letter and the check cannot see it.

## S2-08 (P5-28) Height Hunt  1 star

**Topic: Area of a triangle: the base and its height**

Syllabus outcomes: Area of triangle: base, height and area. Notes on the tag: area of a triangle: base and height (the height is at a right angle to the base)
Puzzle type: Lettered choices (figures). How the pupil checks: Word from the letters of the right choices (6 letters).
**Answers:** 1) HF (base GU); 2) KG (base JM); 3) NU (base LV); 4) JT (base WQ); 5) SQ (base FE); 6) PG (base QH)
**Check result:** Word: COUSIN
**Teacher note:** Six triangles, each with a named base. One dashed line from the opposite vertex meets the base line at a right angle (it is the height); a second dashed line and a side of the triangle do not. Some bases are tilted and some heights fall outside the triangle (the base is extended with a dotted line). The figures are drawn to scale, so a pupil can check a right angle with the corner of a sheet of paper or a set square. The wrong choices are the slanted line and a side. Each choice has a letter and the right choices spell the word. Every other way of choosing was tested against a very large word list. A box left blank gives no letter and the check cannot see it.

## S2-09 (P5-29) Triangle Areas  1 star

**Topic: Area of a triangle: base, height and area**

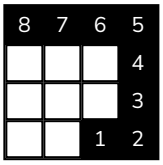
Syllabus outcomes: Area of triangle: base, height and area. Notes on the tag: area of a triangle from its base and height
Puzzle type: Code-breaker (areas of triangles). How the pupil checks: Code word (6 letters).
**Answers:** 1) 20 cm² (base 8, height 5); 2) 18 cm² (base 9, height 4); 3) 21 cm² (base 7, height 6); 4) 55 cm² (base 10, height 11); 5) 72 cm² (base 16, height 9); 6) 30 cm² (base 12, height 5)
**Check result:** Code word: SHRIMP
**Teacher note:** Six triangles with the base and the height (dashed, at a right angle to the base) labelled in cm; some figures have a length on a slanted side that is not needed, and some have the height outside the triangle. The decoy letters stand for the usual slips: forgetting to halve, adding the base and the height, or using the slanted side instead of the height. A box left blank gives no letter and the check cannot see it. Figures are not drawn to scale.

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## S2-10 (P5-30) Triangle Route  2 stars

**Topic: Area of a triangle: base, height and area**

Syllabus outcomes: Area of triangle: base, height and area. Notes on the tag: area of a triangle from its base and height
Puzzle type: Maze / path. How the pupil checks: Route: the line must reach FINISH through the right number of boxes, every box following the rule.
**Answers:** Route (row, column from top-left): (4,3) 12 x 5 / 2 = 30 -> (4,4) 5 x 12 / 2 = 30 -> (3,4) 6 x 10 / 2 = 30 -> (2,4) 10 x 6 / 2 = 30 -> (1,4) 6 x 10 / 2 = 30 -> (1,3) 6 x 10 / 2 = 30 -> (1,2) 10 x 6 / 2 = 30 -> (1,1) 5 x 12 / 2 = 30
**Check result:** Route reaches FINISH through 8 boxes; every box on it follows the rule.




**Teacher note:** A 4 by 4 grid of sixteen triangles, each with its base and its height (dashed) labelled in cm. The boxes on the route have area 30 cm²; some have a length on a slanted side that is not needed. The boxes off the route have a product base x height that gives the area without halving, or an area a little more or less than 30 cm²; some extra boxes that do have area 30 cm² are dead ends. Counting the boxes shows a line that takes a shortcut. Figures are not drawn to scale.

## S2-11 (P5-31) Composite Figures  2 stars

**Topic: Area of composite figures: rectangles and triangles**

Syllabus outcomes: Area of triangle: composite figures, Area of triangle: base, height and area. Notes on the tag: area of composite figures made of rectangles and triangles
Puzzle type: Code-breaker (composite areas). How the pupil checks: Code word (6 letters).
**Answers:** 1) 48 cm² (side); 2) 20 cm² (notch); 3) 36 cm² (house); 4) 32 cm² (notch); 5) 68 cm² (side); 6) 27 cm² (house)
**Check result:** Code word: TEACUP
**Teacher note:** Six figures: a rectangle with a triangle on top (two figures), a rectangle with a triangular notch cut from its top edge (two figures) and a rectangle with a right-angled triangle joined to one side (two figures). The pupil finds the area of the rectangle and of the triangle and adds or subtracts. The decoy letters stand for the usual slips: forgetting to halve the triangle, or adding where the notch must be taken away. Dashed lines are not edges of the figure. Figures are not drawn to scale. A box left blank gives no letter and the check cannot see it.

## S2-12 (P5-32) Count the Cubes  1 star

**Topic: Volume: building cuboids with unit cubes**

Syllabus outcomes: Volume 2.1, Volume 2.3, Volume 2.4. Notes on the tag: building cubes and cuboids with unit cubes; reading drawings of cuboids on an isometric grid; volume of a cuboid
Puzzle type: Code-breaker (cubes in a cuboid). How the pupil checks: Code word (6 letters).
**Answers:** 1) 4 x 2 x 2 = 16; 2) 3 x 1 x 3 = 9; 3) 2 x 2 x 2 = 8; 4) 1 x 3 x 2 = 6; 5) 1 x 4 x 1 = 4; 6) 1 x 2 x 1 = 2
**Check result:** Code word: BANDIT
**Teacher note:** Six cuboids built of unit cubes, drawn from above in isometric view with every unit face visible on the top and two sides. The pupil counts the cubes in one layer and the layers, or multiplies the three edges. The decoy letters stand for the usual slips: counting only the faces that can be seen, adding the three edges, or leaving a layer out. Drawing a cuboid on an isometric grid cannot be checked on a printed page: this page practises reading such a drawing. A box left blank gives no letter and the check cannot see it.

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## S2-13 (P5-33) Cuboid Volumes  1 star

**Topic: Volume of a cube and a cuboid; cm³ and m³**

Syllabus outcomes: Volume 2.4, Volume: cubic units (cm³, m³). Notes on the tag: volume of a cube or cuboid in cubic units (cm³ and m³)
Puzzle type: Code-breaker (volumes). How the pupil checks: Code word (6 letters).
**Answers:** 1) 240 cm³; 2) 45 cm³; 3) 144 m³; 4) 175 cm³; 5) 343 cm³; 6) 147 m³
**Check result:** Code word: CURFEW
**Teacher note:** Six solids with their edges labelled: four cuboids and one cube in cm (the cube has the same edge on all three labels) and, on two figures, in m. The pupil multiplies the three edges; the unit of the answer is cm³ or m³ (the page prints it; there is no conversion between them). The decoy letters stand for the usual slips: adding the edges, the area of a face, or leaving one edge out. Figures are not drawn to scale. A box left blank gives no letter and the check cannot see it.

## S2-14 (P5-34) Tank Match  2 stars

**Topic: Volume: litres and cubic centimetres**

Syllabus outcomes: Volume 2.4, Volume 2.6. Notes on the tag: volume of a cuboid; the relationship between litres and cm³ (1 ℓ = 1000 cm³)
Puzzle type: Matching (lettered cards). How the pupil checks: Letters of the matching cards, tile 1 to 6, spell a word.
**Answers:** F: 25 x 40 x 12 = 12000 cm³ = 12 ℓ; A: 25 x 20 x 30 = 15000 cm³ = 15 ℓ; T: 80 x 20 x 12 = 19200 cm³ = 19.2 ℓ; H: 25 x 20 x 40 = 20000 cm³ = 20 ℓ; O: 50 x 15 x 30 = 22500 cm³ = 22.5 ℓ; M: 80 x 40 x 15 = 48000 cm³ = 48 ℓ
**Check result:** Word: FATHOM
**Teacher note:** Six tanks drawn with their length, width and height in cm, and six tiles with a capacity in litres, in order of size. The pupil multiplies the three edges to find the volume in cm³ and changes it to litres (1 ℓ = 1000 cm³). The letters, read from tile 1 to tile 6, spell a word with exactly one arrangement that is a word, so no wrong matching spells a word. Figures are not drawn to scale. A box left blank gives no letter and the check cannot see it.

## S2-15 (P5-35) Tank Tales  2 stars

**Topic: Volume of liquid in a rectangular tank; litres and cm³**

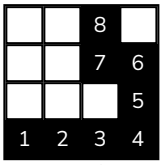
Syllabus outcomes: Volume 2.5, Volume 2.6, Volume 2.4 (problem solving). Notes on the tag: volume of liquid in a rectangular tank; the relationship between ℓ, mℓ and cm³
Puzzle type: Code-breaker (tank stories). How the pupil checks: Code word (6 letters).
**Answers:** 1) 24 ℓ; 2) 62 ℓ; 3) 54 ℓ; 4) 69 ℓ; 5) 672 mℓ; 6) 8 ℓ
**Check result:** Code word: CAMPUS
**Teacher note:** Six tank stories: the capacity of a full tank, the water in a tank filled to a stated depth, the water still needed to fill a tank, the capacity of a small box in mℓ, a cube-shaped tank, and the difference between two tanks. Every step goes forward from the lengths to the volume; no story asks for a missing length or for a water level from a volume. The decoy letters stand for the usual slips: the answer left in cm³ or changed by the wrong power of ten. A box left blank gives no letter and the check cannot see it.

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## S2-16 (P5-36) Angle Route  1 star

**Topic: Geometry: angles on a straight line, at a point and vertically opposite angles**

Syllabus outcomes: Geometry 1.1, Geometry 1.2, Geometry 1.3, Geometry 1.4. Notes on the tag: angles on a straight line, angles at a point, vertically opposite angles, finding unknown angles
Puzzle type: Maze / path. How the pupil checks: Route: the line must reach FINISH through the right number of boxes, every box following the rule.
**Answers:** Route (row, column from top-left): (4,1) ? = 70° -> (4,2) ? = 70° -> (4,3) ? = 70° -> (4,4) ? = 70° -> (3,4) ? = 70° -> (2,4) ? = 70° -> (2,3) ? = 70° -> (1,3) ? = 70°
**Check result:** Route reaches FINISH through 8 boxes; every box on it follows the rule.




**Teacher note:** A 4 by 4 grid of sixteen angle figures: angles on a straight line (they make a straight angle), angles at a point (a full turn) and two straight lines crossing (vertically opposite angles are equal; neighbouring angles make a straight angle). The angle marked ? is 70° on the route. The boxes off the route give an answer a little more or less than 70°, or the angle next to it; some extra boxes that do give 70° are dead ends. Counting the boxes shows a line that takes a shortcut. Figures are not drawn to scale.

## S2-17 (P5-37) Find the Angle  1 star

**Topic: Geometry: finding unknown angles**

Syllabus outcomes: Geometry 1.1, Geometry 1.2, Geometry 1.3, Geometry 1.4. Notes on the tag: angles on a straight line, angles at a point, vertically opposite angles, finding unknown angles
Puzzle type: Lettered choices (figures). How the pupil checks: Word from the letters of the right choices (6 letters).
**Answers:** 1) 60°; 2) 70°; 3) 115°; 4) 40°; 5) 50°; 6) 145°
**Check result:** Word: KNIGHT
**Teacher note:** Six figures, two of each kind: angles on a straight line (a straight angle), angles at a point (a full turn) and two crossing lines (vertically opposite angles equal, neighbours make a straight angle). The wrong choices are the usual slips: the angle next to the right one, a full turn used on a straight line, or the sum of the given angles. Each choice has a letter and the right choices spell the word. Every other way of choosing was tested against a very large word list. Figures are not drawn to scale. A box left blank gives no letter and the check cannot see it.

## S2-18 (P5-38) Triangle Angles  1 star

**Topic: Geometry: angle sum of a triangle; isosceles, equilateral and right-angled triangles**

Syllabus outcomes: Geometry: angle sum of a triangle, Geometry: properties of triangles, Geometry 1.4. Notes on the tag: angle sum of a triangle; properties of isosceles, equilateral and right-angled triangles; finding unknown angles
Puzzle type: Code-breaker (angles in triangles). How the pupil checks: Code word (6 letters).
**Answers:** 1) 78° (sum); 2) 60° (equi); 3) 61° (apex); 4) 98° (base); 5) 101° (sum); 6) 30° (right)
**Check result:** Code word: WIZARD
**Teacher note:** Six triangles: two with two angles given (the three angles of a triangle make a straight angle), an isosceles triangle with the angle at the top given, an isosceles triangle with a base angle given, an equilateral triangle and a right-angled triangle with one other angle given. Equal sides carry tick marks. The decoy letters stand for the usual slips: giving a straight angle minus one angle only, adding the angles, or not halving the base angles. Figures are drawn to scale in these pages but the pupil should use the angle facts, not a protractor. A box left blank gives no letter and the check cannot see it.

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## S2-19 (P5-39) Shape Statements  2 stars

**Topic: Geometry: parallelogram, rhombus and trapezium**

Syllabus outcomes: Geometry: parallelogram, rhombus, trapezium, Geometry 1.4. Notes on the tag: properties of a parallelogram, rhombus and trapezium; their unknown angles
Puzzle type: True or false table (three choices). How the pupil checks: Word from the letters of the right choices (6 letters).
**Answers:** 1) F (KLMN is a trapezium. KL is parallel to NM. ∠N = 79°. ∠K = 79°.); 2) N (EFGH is a trapezium. EF is parallel to HG. ∠F = 54°. ∠H = 54°.); 3) T (ABCD is a rhombus. ∠D = 79°. ∠A = 101°.); 4) T (TUVW is a trapezium. TU is parallel to WV. ∠U = 61°. ∠V = 119°.); 5) F (WXYZ is a parallelogram. ∠W = 62°. ∠X = 62°.); 6) T (PQRS is a parallelogram. ∠Q = 77°. ∠S = 77°.)
**Check result:** Word: CANOPY
**Teacher note:** Six statements about a parallelogram, a rhombus or a trapezium with two angles given. In a parallelogram or rhombus opposite angles are equal and neighbouring angles make a straight angle; in a trapezium the two angles on the same non-parallel side make a straight angle, and two angles that are not on the same side of the trapezium cannot be found from each other, so the answer is N. Each choice (T, F, N) has a letter and the right choices spell the word. Every other way of choosing was tested against a very large word list. A box left blank gives no letter and the check cannot see it.

## S2-20 (P5-40) Angle Chains  2 stars

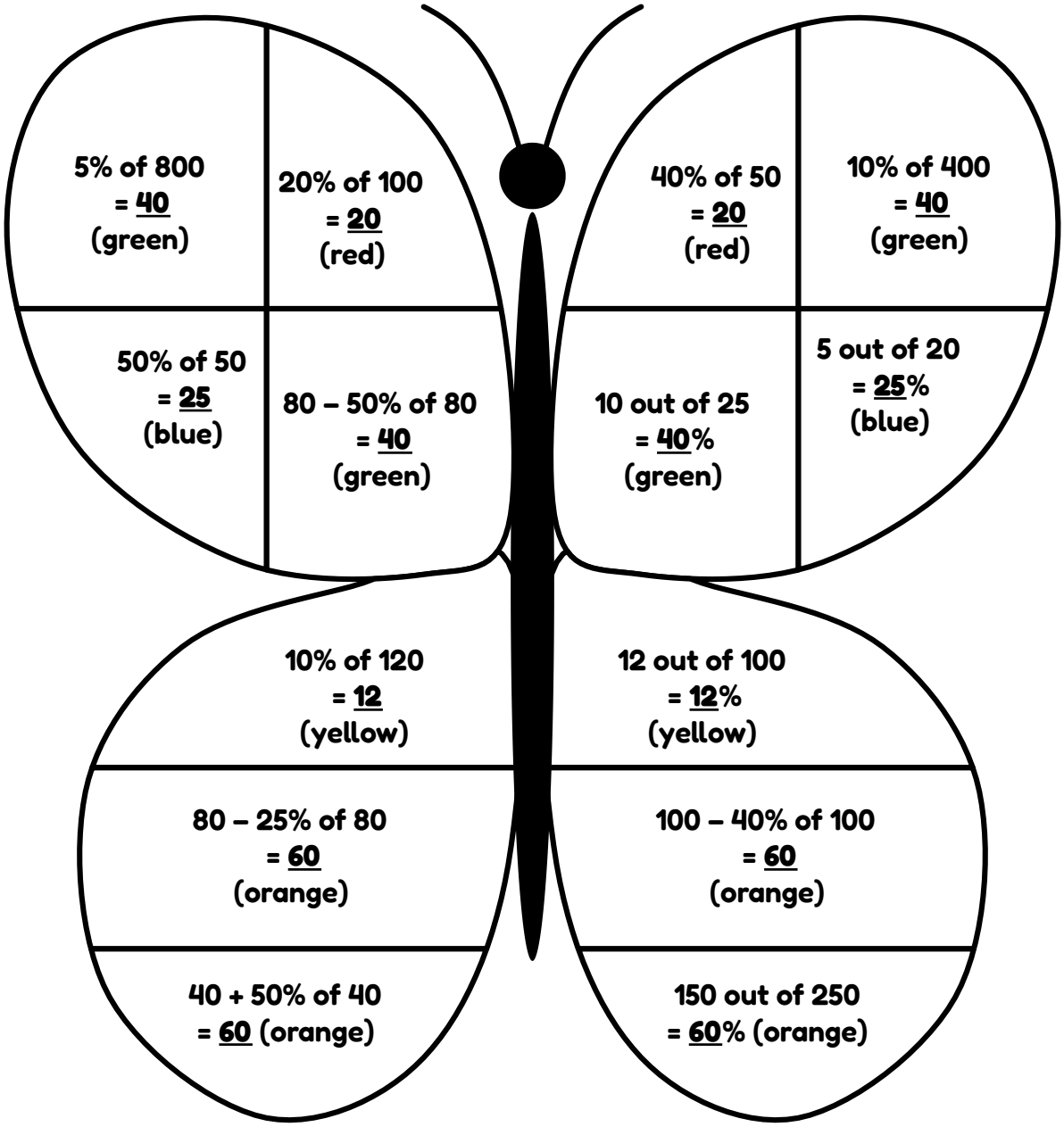
**Topic: Geometry: unknown angles in two steps**

Syllabus outcomes: Geometry 1.4, Geometry: angle sum of a triangle, Geometry 1.1, Geometry 1.3, Geometry: properties of triangles (problem solving). Notes on the tag: finding unknown angles in two steps: angle sum of a triangle with angles on a straight line or vertically opposite angles
Puzzle type: Code-breaker (angle chains). How the pupil checks: Code word (6 letters).
**Answers:** 1) 114° (isoext); 2) 56° (ext2); 3) 73° (ext2); 4) 109° (ext); 5) 147° (isoext); 6) 136° (ext)
**Check result:** Code word: BOTANY
**Teacher note:** Six two-step figures, two of each kind: a triangle with two angles given and one side extended (find the third angle, then the angle on the straight line); an isosceles triangle with the top angle given and the base extended (find the base angle, then the angle on the straight line); and a triangle on a straight line with the two outside angles given (find the two inside angles on the line, then the third angle of the triangle). The decoy letters stand for the usual slips: the angle inside the triangle instead of the angle on the line, or stopping after the first step. Figures are not drawn to scale. A box left blank gives no letter and the check cannot see it.

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# Answers: S2-02 (P5-22) Colour the Butterfly

Every part shows the question, the answer underlined and the colour in brackets. Pupils colour their own copy. Colour key: 12 = yellow, 20 = red, 25 = blue, 40 = green, 60 = orange.




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